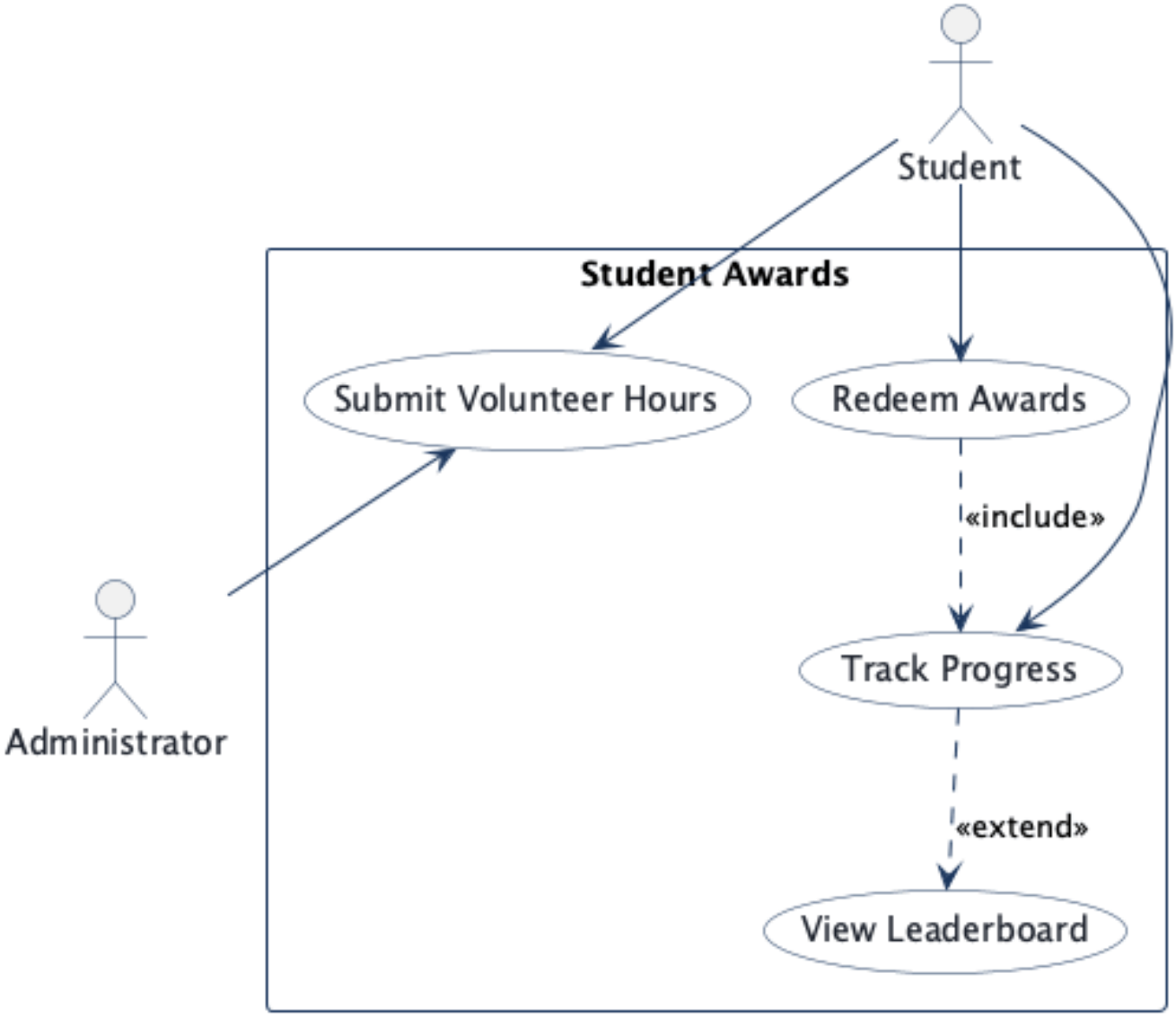
COMP 3613 Assignment 2

Individual report

Field	Value
Student name	Dana Ramtahal
Student ID	816047190
Deployed application	https://comp3613a1-pln2.onrender.com
GitHub repository	https://github.com/DanaRPR/comp3613a1

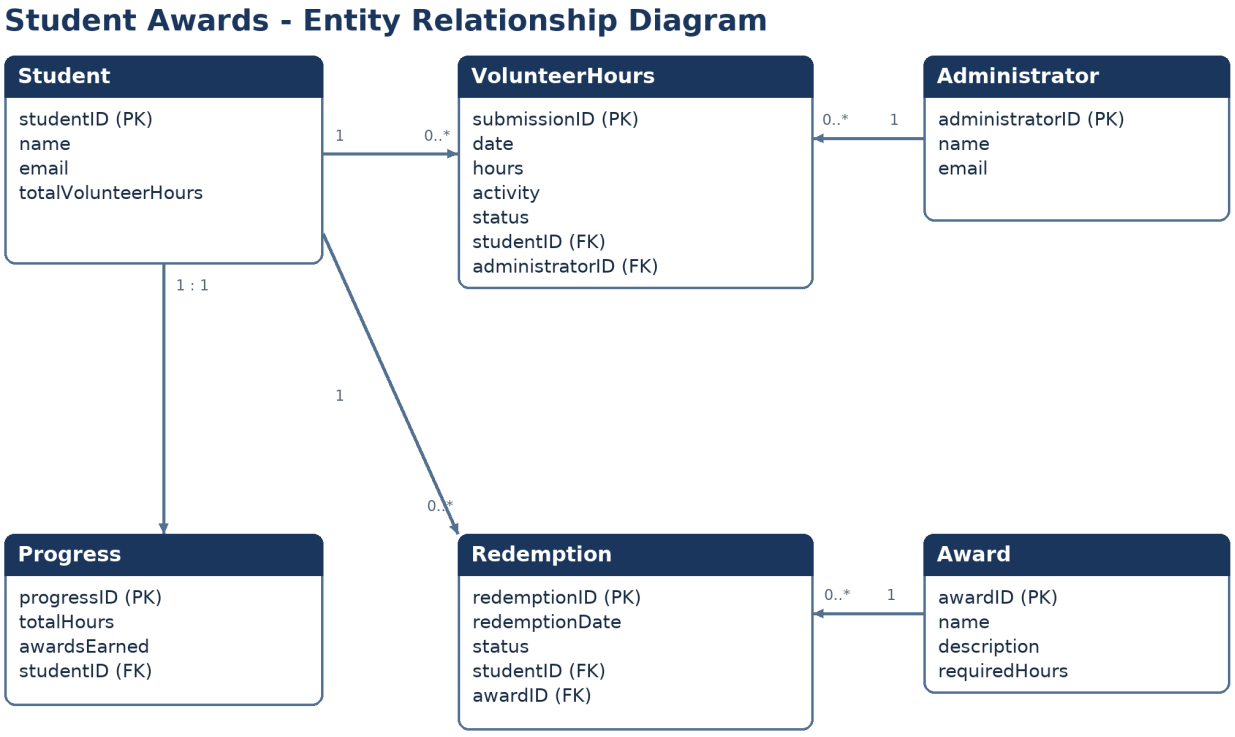
Project Description

Student Awards is a web-based incentive system for student volunteering. Students submit volunteer hours for administrator review, track approved hours and leaderboard rankings, and redeem awards when they meet the required-hour thresholds. Administrators approve or reject pending submissions; approved hours update student totals and progress. The application uses FastAPI, SQLModel, server-rendered HTML views, and a layered route/service/repository/model structure.



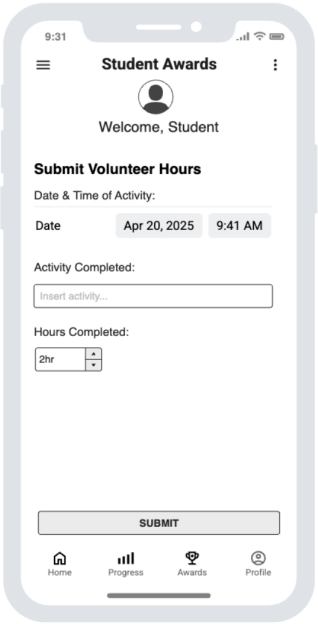
Use Case Diagram

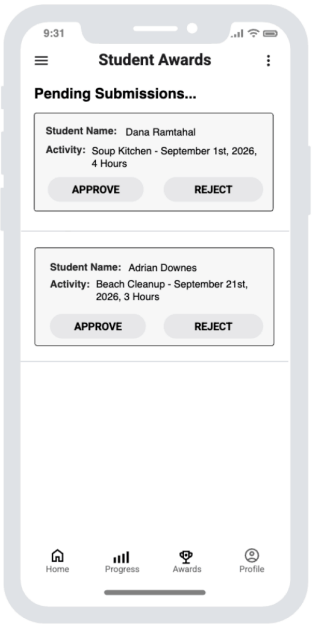
Students submit volunteer hours, track progress, view the leaderboard, and redeem awards. Administrators review submissions by approving or rejecting them. Viewing the leaderboard is optional after tracking progress; redeeming an award includes checking progress.



Entity Relationship Diagram

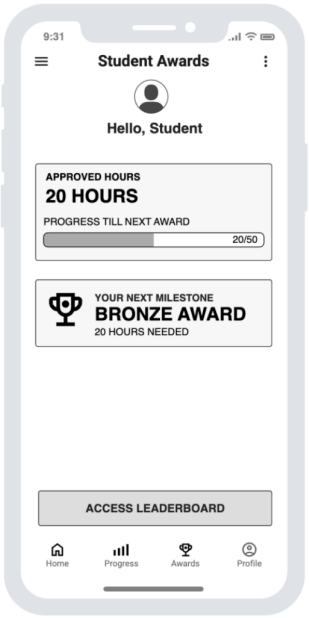
SUBMIT VOLUNTEER HOURS:

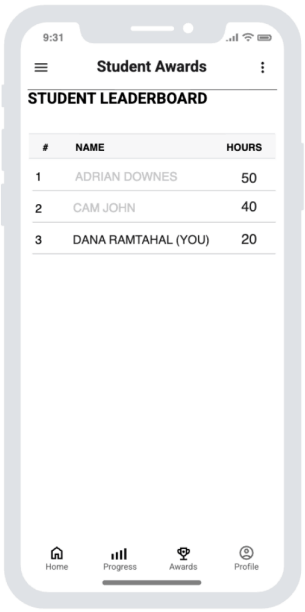




Application Wireframe - Submit Volunteer Hours

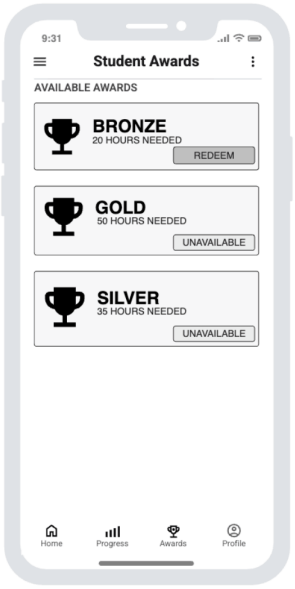
TRACK PROGRESS AND VIEW LEADERBAORD:

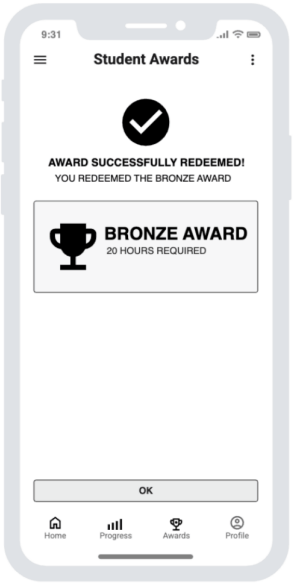




Application Wireframe - Track Progress and View Leaderboard

REDEEM AWARDS:





Application Wireframe - Redeem Awards

Test Plans

User Acceptance Test Plan and Completed Results

The scenarios are based on the application wireframes. The actual results and PASS outcomes below are the recorded user acceptance observations.

Test ID	Scenario	Role	Expected Result	Actual Result	Status
UAT-01	Submit Volunteer Hours	Student	Success message appears; new submission appears in administrator pending list.	Student submitted 5 hours for a community clean-up and received a success message.	PASS
UAT-02	Administrator Approves Volunteer Hours	Administrator, then Student	Submission leaves pending list; approved hours increase by the submitted amount.	Administrator approved the 5-hour Community clean-up submission; it disappeared from the pending list.	PASS
UAT-03	Reject a Pending Submission	Administrator, then Student	Submission leaves pending list; approved-hours total does not increase because of rejection.	Administrator rejected the pending submission; it disappeared from the pending list, leaving no pending submissions.	PASS
UAT-04	View Progress and Next Milestone	Student	Page shows current approved hours and, when applicable, the next unredeemed award milestone.	Student total increased to 10 hours after approval of the 5-hour Community clean-up submission.	PASS
UAT-05	View Leaderboard	Student	Students are ordered by approved hours; administrator is excluded; signed-in student is marked.	Leaderboard showed bob in first place with 10 hours; other students with 0 hours appeared below.	PASS
UAT-06	Redeem an Eligible Award	Student	Eligible award can be redeemed and is then displayed as redeemed.	Student reached 20 approved hours; Bronze became eligible. After clicking Redeem, the application displayed “Redeemed”.	PASS
UAT-07	Attempt to Redeem an Unavailable Award	Student	An ineligible or already redeemed award cannot be redeemed.	With 10 approved hours, Bronze, Silver and Gold were unavailable and redemption buttons were disabled.	PASS

All seven UAT scenarios were recorded as PASS. UAT observations are separate from the automated unit and integration test evidence.

Unit Test Plan

Unit tests use fake repositories, session objects, templates, and services. They do not use the SQLite integration fixture or connect to the configured application database. The table describes the assertions made by the current test functions.

Test ID	Test Function	Verification / Expected Behaviour	Result
UT-01	test_submit_hours_adds_pending_submission()	Checks pending status, student ID, hours, activity, and that the fake repository stores the returned submission; date is not asserted.	PASS
UT-02	test_approve_updates_submission_and_totals()	Checks approved status, administrator ID, student total 10→18, progress total 15→23, and fake-session add; commit/persisted state is not asserted.	PASS
UT-03	test_approve_raises_value_error_for_missing_submission()	Expects ValueError matching “Submission not found”.	PASS
UT-04	test_reject_marks_submission_rejected()	Checks rejected status, administrator ID, and fake-session add; unchanged totals are not asserted.	PASS
UT-05	test_get_available_awards_marks_eligible_and_not_redeemed()	Checks an above-threshold award is available and not already redeemed.	PASS
UT-06	test_get_available_awards_marks_redeemed_and_ineligible()	Checks an under-threshold award that is also listed as redeemed is unavailable and already redeemed; conditions are combined.	PASS
UT-07	test_redeem_raises_for_missing_award()	Expects ValueError matching “Award not found”.	PASS
UT-08	test_redeem_raises_for_duplicate_redemption()	Expects ValueError matching “already redeemed” when the award ID is already redeemed.	PASS
UT-09	test_get_leaderboard_assigns_rank_and_current_student()	Checks first-row rank, name, hours, current-student flag, and second row’s non-current flag; fake rows are already ordered.	PASS
UT-10	test_get_progress_returns_next_award_and_threshold()	Checks returned progress object, next unredeemed award, and threshold of 25.	PASS
UT-11	test_submit_action_renders_value_error_message()	Fake service raises ValueError(“invalid submission”); checks HTTP 400, submit_hours.html template, and error context text.	PASS

Execution result: 11 unit tests passed, 0 failed.

Integration Test Plan

Integration tests use SQLModel records and the shared session fixture, which creates a temporary SQLite database under tmp_path. They do not connect to the deployed database. The table describes the assertions made by the current test functions.

Test ID	Test Function	Verification / Expected Behaviour	Result
IT-01	test_submit_hours_creates_pending_submission()	Checks created record exists, is pending, and has the expected student ID and 5 hours; date/activity are not asserted.	PASS
IT-02	test_approve_submission_updates_student_and_progress()	Checks approved status, stored student total of 8, progress total of 8, and zero awards earned; administrator ID is not asserted.	PASS
IT-03	test_approve_submission_creates_progress_when_missing()	Checks approval and creation of a Progress row with 8 hours and zero awards earned.	PASS
IT-04	test_approve_submission_rejects_duplicate_review()	Checks a second approval raises ValueError matching “already been reviewed”; post-error totals are not checked.	PASS
IT-05	test_approve_missing_submission_raises_value_error()	Expects ValueError matching “Submission not found”.	PASS
IT-06	test_reject_submission_sets_rejected_status()	Checks rejected status and administrator ID 7; unchanged totals are not asserted.	PASS
IT-07	test_reject_missing_submission_raises_value_error()	Expects ValueError matching “Submission not found”.	PASS
IT-08	test_pending_submissions_include_student_names()	Checks two rows, expected student names, and pending activities; does not directly assert the reviewed row’s absence by ID/status.	PASS
IT-09	test_get_available_awards_eligible_and_unredeemed()	Checks an above-threshold, unredeemed award is available and not already redeemed.	PASS

Integration Test Plan (continued)

Remaining integration test cases.

Test ID	Test Function	Verification / Expected Behaviour	Result
IT-10	test_get_available_awards_marks_ineligible_and_redeemed()	Checks an under-threshold award that is also redeemed is unavailable and already redeemed; conditions are combined.	PASS
IT-11	test_redeem_award_creates_redemption_and_updates_progress()	Checks returned redemption status/student ID and persisted Progress.awardsEarned == 1; no independent query for the redemption row.	PASS
IT-12	test_redeem_award_does_not_create_missing_progress()	Checks returned redemption is completed and no Progress row exists afterward; no independent query for a saved Redemption row.	PASS
IT-13	test_redeem_award_rejects_insufficient_hours()	Expects ValueError matching “You need”; database state afterward is not checked.	PASS
IT-14	test_redeem_award_rejects_duplicate()	Checks a second redemption raises ValueError matching “already redeemed”; redemption-row count is not checked.	PASS
IT-15	test_leaderboard_excludes_admins_and_orders_students_by_hours()	Checks returned names/order, first-row rank, and current-student flag; seeded order excludes admin, but not all displayed hour values or tie handling.	PASS
IT-16	test_get_progress_returns_next_unredeemed_award()	Checks progress total 30 and that next award and threshold are None when the only award is redeemed.	PASS
IT-17	test_get_progress_returns_first_unredeemed_award()	Checks Silver is selected after Bronze is redeemed and threshold is 25; returned progress object/total are not asserted.	PASS

Execution result: 17 integration tests passed, 0 failed. Together with the 11 unit tests, the automated suite completed with 28 passed and 0 failed.

• **Postman collection:** student-awards-api-tests.postman_collection.json, exported as a Postman Collection v2.1 JSON file with 21 requests. It covers login and registration, student and administrator pages/actions, form validation, and access-control responses.
• **Postman run evidence:** The saved collection-run screenshots show 24 passed, 0 failed, and 0 errors. Postman was not rerun for this PDF revision.

```
(.venv) danaramtahal@Danas-Macbook comp3613a1 % PYTHONPATH=. pytest tests/unit -v
================================== test session starts ==================================
platform darwin -- Python 3.14.0, pytest-9.1.1, pluggy-1.6.0 -- /Users/danaramtahal/comp3613a1/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/danaramtahal/comp3613a1
configfile: pyproject.toml
plugins: asyncio-1.4.0, anyio-4.15.1
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 11 items

tests/unit/test_award_service_unit.py::test_get_available_awards_marks_eligible_and_not_redeemed PASSED [  9%]
tests/unit/test_award_service_unit.py::test_get_available_awards_marks_redeemed_and_ineligible PASSED [ 18%]
tests/unit/test_award_service_unit.py::test_redeem_raises_for_missing_award PASSED      [ 27%]
tests/unit/test_award_service_unit.py::test_redeem_raises_for_duplicate_redemption PASSED [ 36%]
tests/unit/test_leaderboard_service_unit.py::test_get_leaderboard_assigns_rank_and_current_student PASSED [ 45%]
tests/unit/test_leaderboard_service_unit.py::test_get_progress_returns_next_award_and_threshold PASSED [ 54%]
tests/unit/test_submit_hours_router_unit.py::test_submit_action_renders_value_error_message PASSED [ 63%]
tests/unit/test_volunteer_hours_service_unit.py::test_submit_hours_adds_pending_submission PASSED [ 72%]
tests/unit/test_volunteer_hours_service_unit.py::test_approve_updates_submission_and_totals PASSED [ 81%]
tests/unit/test_volunteer_hours_service_unit.py::test_approve_raises_value_error_for_missing_submission PASSED [ 90%]
tests/unit/test_volunteer_hours_service_unit.py::test_reject_marks_submission_rejected PASSED [100%]

================================== 11 passed in 0.16s ===================================
(.venv) danaramtahal@Danas-Macbook comp3613a1 %
```

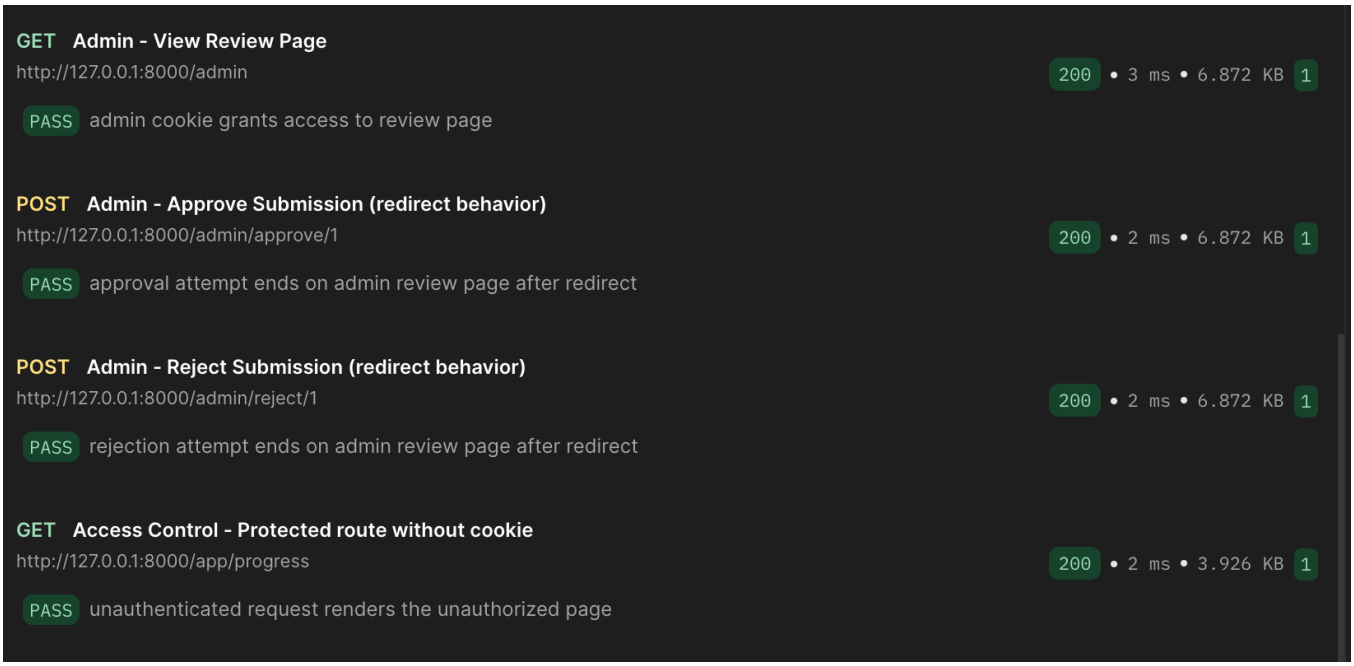
Unit Test Report

```
(.venv) danaramtahal@Danas-Macbook comp3613a1 % PYTHONPATH=. pytest tests/integration -v
============================================== test session starts ==============================================
platform darwin -- Python 3.14.0, pytest-9.1.1, pluggy-1.6.0 -- /Users/danaramtahal/comp3613a1/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/danaramtahal/comp3613a1
configfile: pyproject.toml
plugins: asyncio-1.4.0, anyio-4.15.1
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 17 items

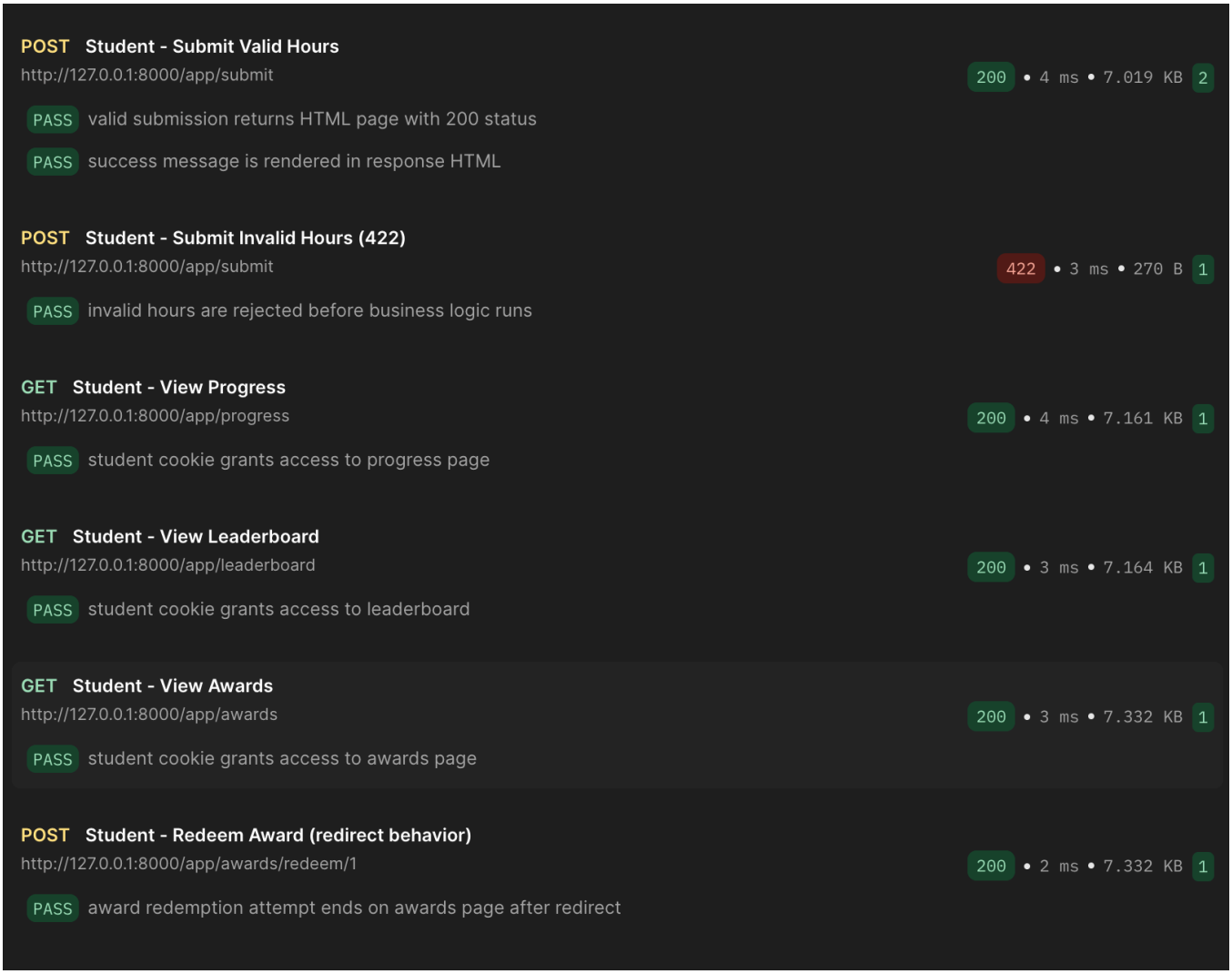
tests/integration/test_award_service_integration.py::test_get_available_awards_eligible_and_unredeemed PASSED                  [  5%]
tests/integration/test_award_service_integration.py::test_get_available_awards_marks_ineligible_and_redeemed PASSED            [ 11%]
tests/integration/test_award_service_integration.py::test_redeem_award_creates_redemption_and_updates_progress PASSED          [ 17%]
tests/integration/test_award_service_integration.py::test_redeem_award_does_not_create_missing_progress PASSED                 [ 23%]
tests/integration/test_award_service_integration.py::test_redeem_award_rejects_insufficient_hours PASSED                       [ 29%]
tests/integration/test_award_service_integration.py::test_redeem_award_rejects_duplicate PASSED                                [ 35%]
tests/integration/test_leaderboard_service_integration.py::test_leaderboard_excludes_admins_and_orders_students_by_hours PASSED [ 41%]
tests/integration/test_leaderboard_service_integration.py::test_get_progress_returns_next_unredeemed_award PASSED             [ 47%]
tests/integration/test_leaderboard_service_integration.py::test_get_progress_returns_first_unredeemed_award PASSED            [ 52%]
tests/integration/test_volunteer_hours_service_integration.py::test_submit_hours_creates_pending_submission PASSED            [ 58%]
tests/integration/test_volunteer_hours_service_integration.py::test_approve_submission_updates_student_and_progress PASSED    [ 64%]
tests/integration/test_volunteer_hours_service_integration.py::test_approve_submission_creates_progress_when_missing PASSED   [ 70%]
tests/integration/test_volunteer_hours_service_integration.py::test_approve_submission_rejects_duplicate_review PASSED        [ 76%]
tests/integration/test_volunteer_hours_service_integration.py::test_approve_missing_submission_raises_value_error PASSED      [ 82%]
tests/integration/test_volunteer_hours_service_integration.py::test_reject_submission_sets_rejected_status PASSED             [ 88%]
tests/integration/test_volunteer_hours_service_integration.py::test_reject_missing_submission_raises_value_error PASSED      [ 94%]
tests/integration/test_volunteer_hours_service_integration.py::test_pending_submissions_include_student_names PASSED          [100%]

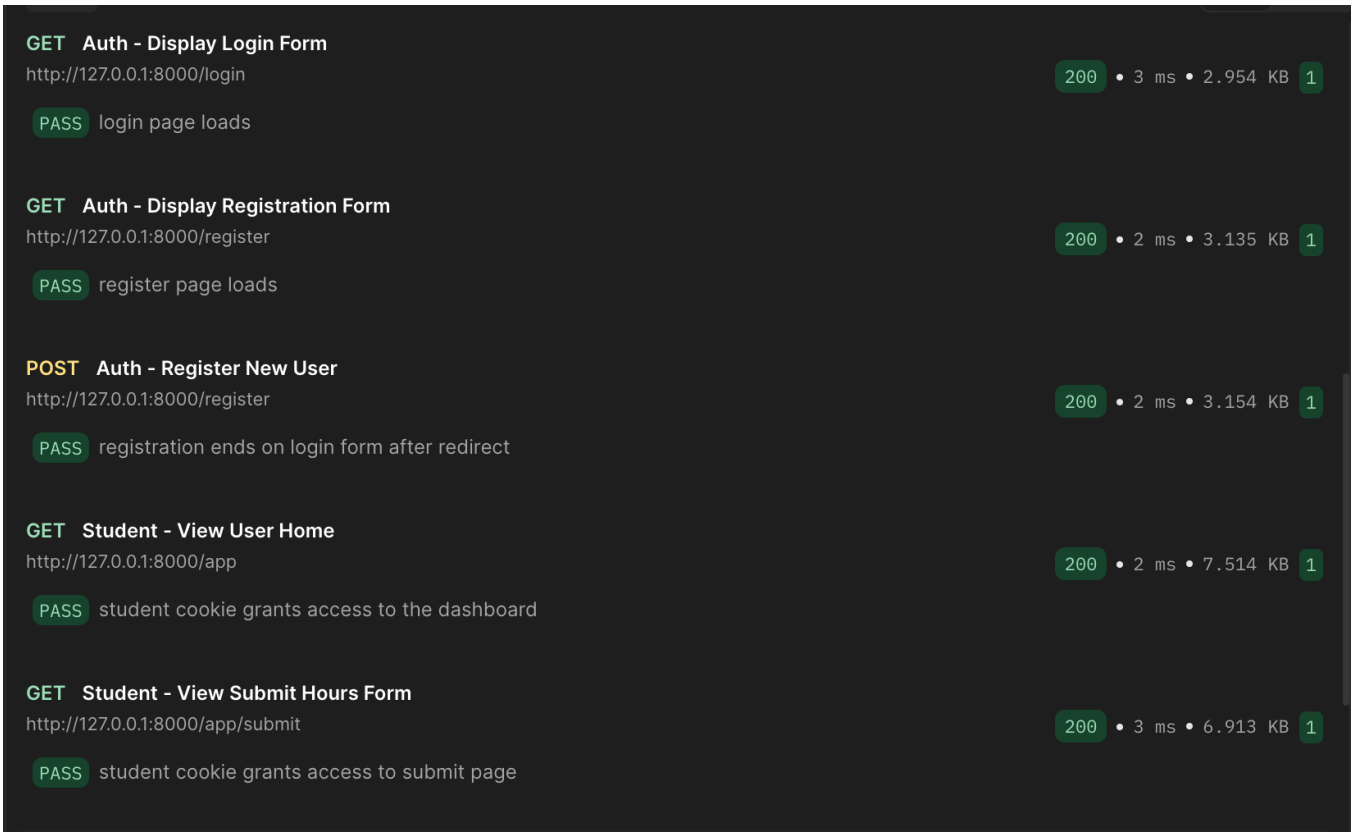
============================================== 17 passed in 0.22s ===============================================
(.venv) danaramtahal@Danas-Macbook comp3613a1 %
```

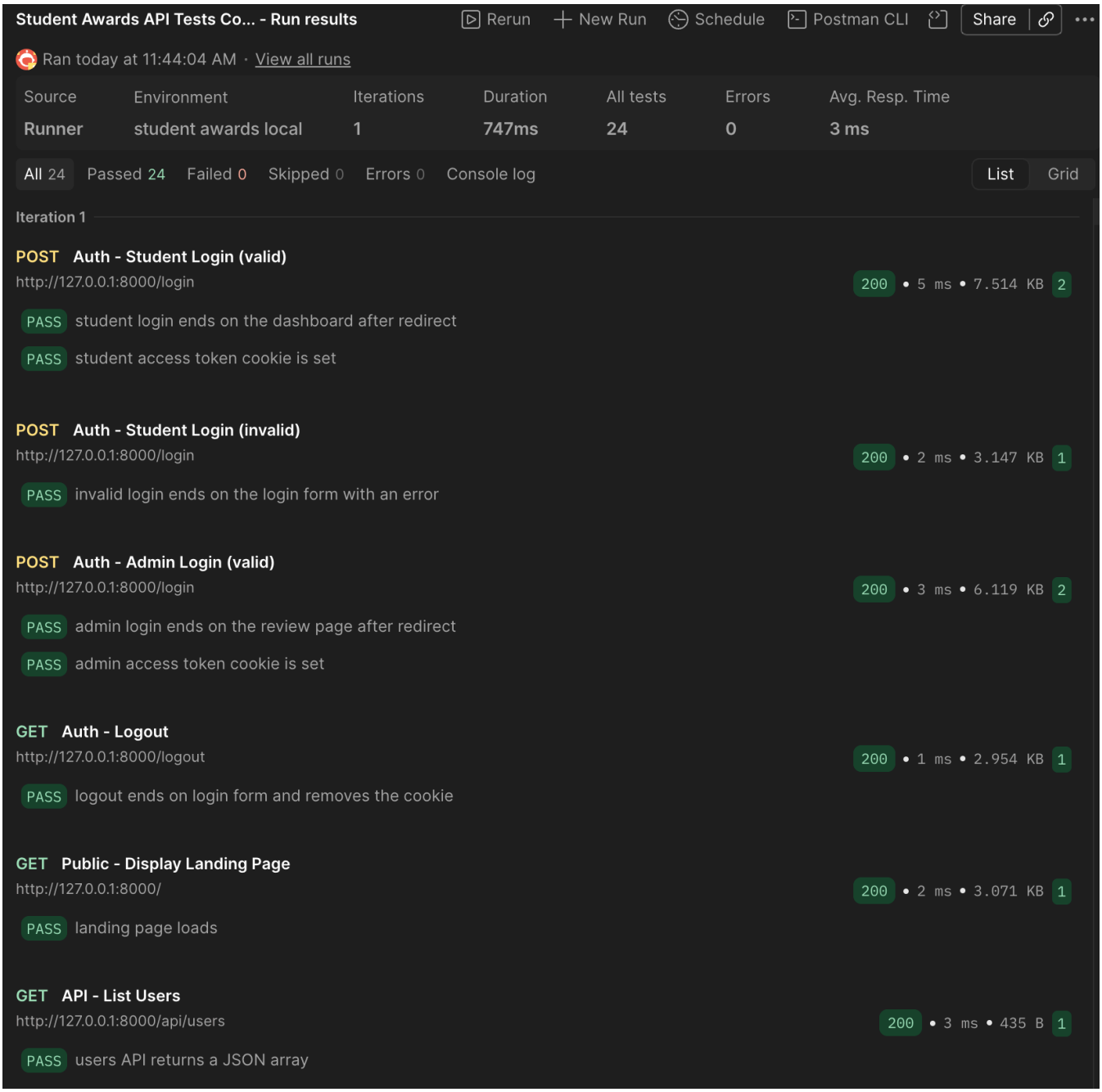
Integration Test Report



Postman Collection Run Report




GET Auth - Display Login Form
http://127.0.0.1:8000/login
200 • 3 ms • 2.954 KB 1
PASS login page loads
GET Auth - Display Registration Form
http://127.0.0.1:8000/register
200 • 2 ms • 3.135 KB 1
PASS register page loads
POST Auth - Register New User
http://127.0.0.1:8000/register
200 • 2 ms • 3.154 KB 1
PASS registration ends on login form after redirect
GET Student - View User Home
http://127.0.0.1:8000/app
200 • 2 ms • 7.514 KB 1
PASS student cookie grants access to the dashboard
GET Student - View Submit Hours Form
http://127.0.0.1:8000/app/submit
200 • 3 ms • 6.913 KB 1
PASS student cookie grants access to submit page


Student Awards API Tests Co... - Run results
Rerun
New Run
Schedule
Postman CLI
Share
Ran today at 11:44:04 AM · View all runs
Source
Environment
Iterations
Duration
All tests
Errors
Avg. Resp. Time
Runner
student awards local
1
747ms
24
0
3 ms
All 24
Passed 24
Failed 0
Skipped 0
Errors 0
Console log
List
Grid
Iteration 1
POST Auth - Student Login (valid)
http://127.0.0.1:8000/login
200 • 5 ms • 7.514 KB 2
PASS student login ends on the dashboard after redirect
PASS student access token cookie is set
POST Auth - Student Login (invalid)
http://127.0.0.1:8000/login
200 • 2 ms • 3.147 KB 1
PASS invalid login ends on the login form with an error
POST Auth - Admin Login (valid)
http://127.0.0.1:8000/login
200 • 3 ms • 6.119 KB 2
PASS admin login ends on the review page after redirect
PASS admin access token cookie is set
GET Auth - Logout
http://127.0.0.1:8000/logout
200 • 1 ms • 2.954 KB 1
PASS logout ends on login form and removes the cookie
GET Public - Display Landing Page
http://127.0.0.1:8000/
200 • 2 ms • 3.071 KB 1
PASS landing page loads
GET API - List Users
http://127.0.0.1:8000/api/users
200 • 3 ms • 435 B 1
PASS users API returns a JSON array